

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

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COURSE NAME: Computer Networks

COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

- 1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.**

Aim

To design and implement a simple Local Area Network (LAN) consisting of two PCs connected through a switch using Cisco Packet Tracer, configure IP addresses, verify connectivity using the **ping** command, capture packets using Wireshark, identify Ethernet frame fields, and explain how the Data Link Layer uses MAC addressing for reliable frame delivery.

Objectives

- Design a LAN topology using Cisco Packet Tracer.
- Configure IPv4 addresses on end devices.
- Test network connectivity using the ping command.
- Capture ICMP packets using Wireshark.
- Analyze Ethernet frame fields.
- Understand the role of the Data Link Layer in local communication.

Software and Hardware Requirements

- Cisco Packet Tracer
- Wireshark
- PC/Laptop
- Cisco 2960 Switch
- Two PCs

- Copper Straight-Through Cables

Theory

A Local Area Network (LAN) connects computers within a limited geographical area such as a laboratory or office. In Ethernet LANs, communication takes place at the Data Link Layer (Layer 2) using MAC addresses, while logical addressing is handled by the Network Layer (Layer 3) using IP addresses.

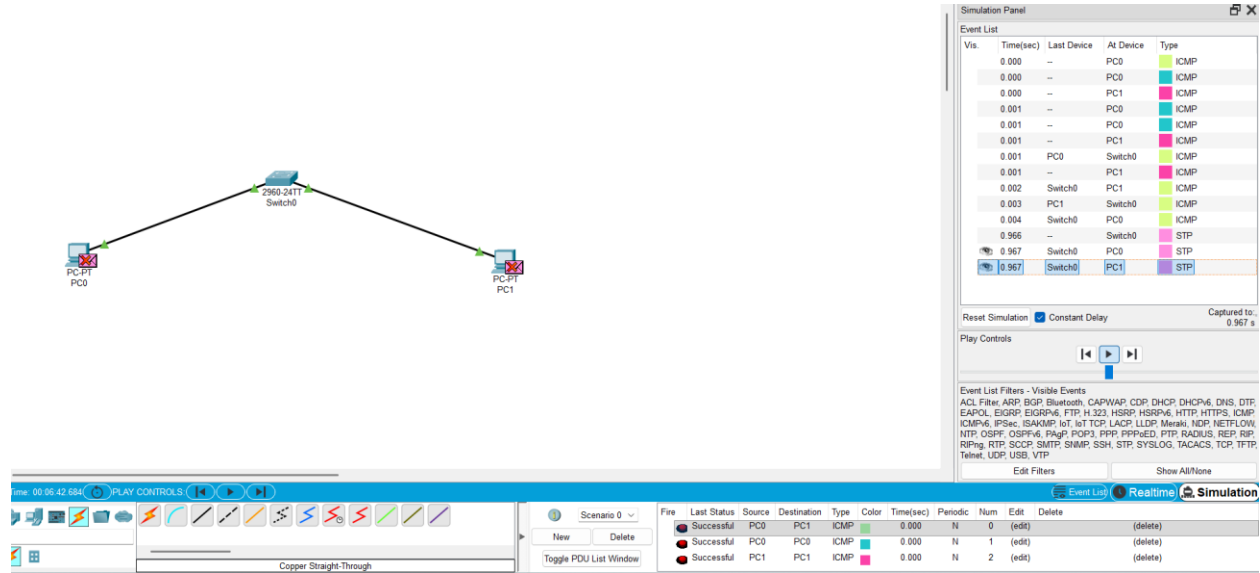
Before communication begins, the sender resolves the destination IP address to its MAC address using the Address Resolution Protocol (ARP). The Ethernet frame is then created with the source and destination MAC addresses. The switch reads the destination MAC address and forwards the frame only to the correct port. The receiver verifies the Frame Check Sequence (FCS) to ensure that the frame has not been corrupted during transmission.

This process enables reliable communication between devices within the same LAN.

Algorithm

1. Create a LAN topology with two PCs and one switch.
2. Connect devices using Ethernet cables.
3. Configure IPv4 addresses.
4. Verify network connectivity using the ping command.
5. Start packet capture in Wireshark.
6. Generate ICMP packets.
7. Stop the capture.
8. Analyze Ethernet frame fields.
9. Interpret the Data Link Layer operation.
10. Validate successful communication.

OUTPUT:



RESULT:

The ping was successful, confirming communication between the two PCs. Wireshark showed the Ethernet frame fields, including the Source MAC Address, Destination MAC Address, EtherType, and FCS.

2. **Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.**

Aim

To create a network topology consisting of two LANs connected through a router using Cisco Packet Tracer, configure IPv4 addressing, subnet masks and default gateways, verify communication using the ping command, capture ICMP packets using Wireshark, and analyze the IPv4 header fields.

Objectives

- Design two LANs connected by a router.
- Configure IPv4 addressing.
- Configure default gateways.
- Verify communication using ping.
- Capture and analyze IPv4 packets.
- Understand the role of the Network Layer.

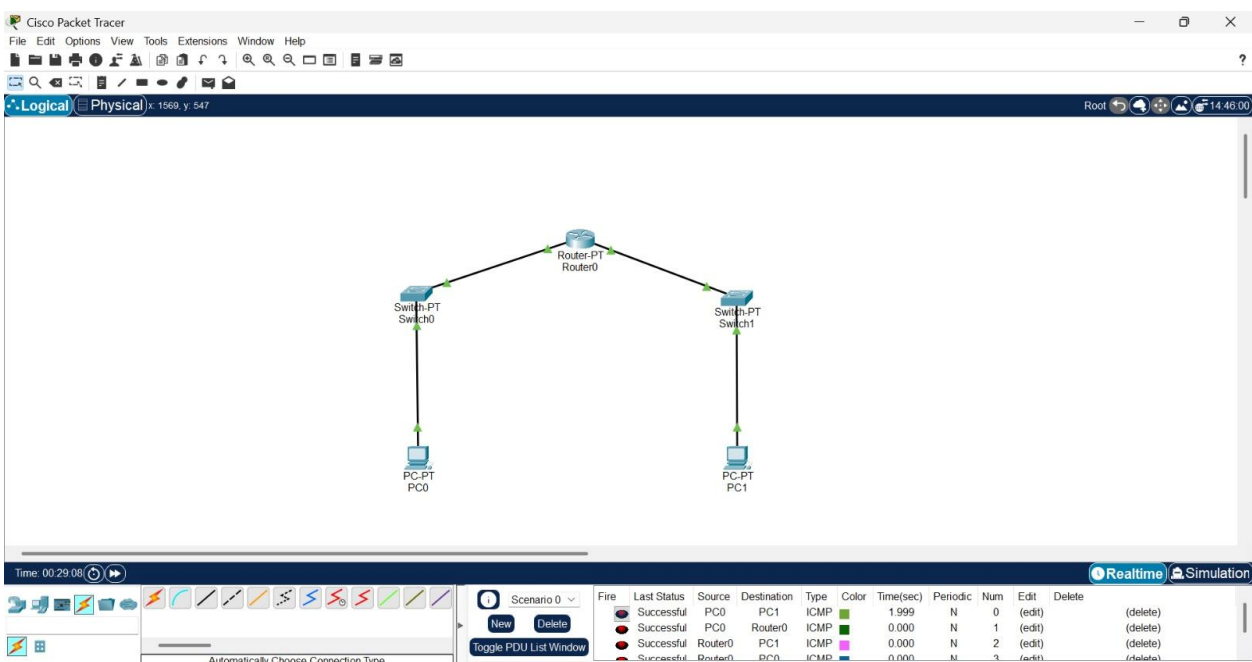
Requirements

- Cisco Packet Tracer
- Wireshark
- One Router (1941/2911)
- One Switch for each LAN
- Four PCs

Algorithm

- 1) Open Cisco Packet Tracer.
- 2) Place one router, two switches, and four PCs in the workspace.
- 3) Connect each PC to its respective switch using Copper Straight-Through cables.
- 4) Connect both switches to the router.
- 5) Configure IPv4 addresses and subnet masks for all PCs.
- 6) Configure IP addresses for the router interfaces and enable them using the no shutdown command.
- 7) Configure the default gateway on each PC.
- 8) Verify network connectivity using the ping command between PCs in different LANs.
- 9) Start packet capture in Wireshark.
- 10) Perform the ping operation and capture the ICMP packets.
- 11) Stop the capture and analyze the IPv4 header fields (Source IP, Destination IP, TTL, Protocol Number, and Header Length).
- 12) Record the observations and verify successful communication.

OUTPUT:



RESULT:

The ping was successful, confirming communication between the two LANs. Wireshark displayed the IPv4 header fields, including the Source IP, Destination IP, TTL, Protocol Number, and Header Length.

3. **Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.**

Aim

To construct a Local Area Network (LAN) with multiple hosts connected through a switch, clear the ARP cache on one host, initiate communication using the ping command, capture ARP packets using Wireshark, and analyze how ARP resolves IP addresses into MAC addresses.

Objectives

- Construct a LAN with multiple hosts.
- Configure IPv4 addresses.
- Clear the ARP cache.
- Capture ARP Request and ARP Reply packets using Wireshark.
- Analyze sender and target MAC/IP addresses.
- Understand the working of the Address Resolution Protocol (ARP).

Requirements

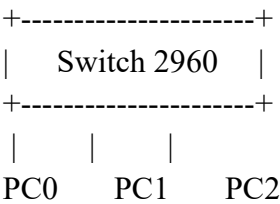
- Cisco Packet Tracer
- Wireshark
- One Cisco 2960 Switch
- Three PCs
- Copper Straight-Through Cables

Theory

The **Address Resolution Protocol (ARP)** is a Network Layer support protocol that maps a logical **IPv4 address** to a physical **MAC address** within a Local Area Network (LAN). Before a device sends data to another device on the same network, it must know the destination MAC address.

If the MAC address is not available in the sender's ARP cache, the sender broadcasts an **ARP Request** asking, "Who has this IP address?" The device with the matching IP address responds with an **ARP Reply** containing its MAC address. The sender stores this information in its ARP cache and uses it to send Ethernet frames directly to the destination.

Network Topology



IP Addressing Table

Device	IP Address	Subnet Mask
PC0	192.168.1.2	255.255.255.0
PC1	192.168.1.3	255.255.255.0
PC2	192.168.1.4	255.255.255.0

Algorithm

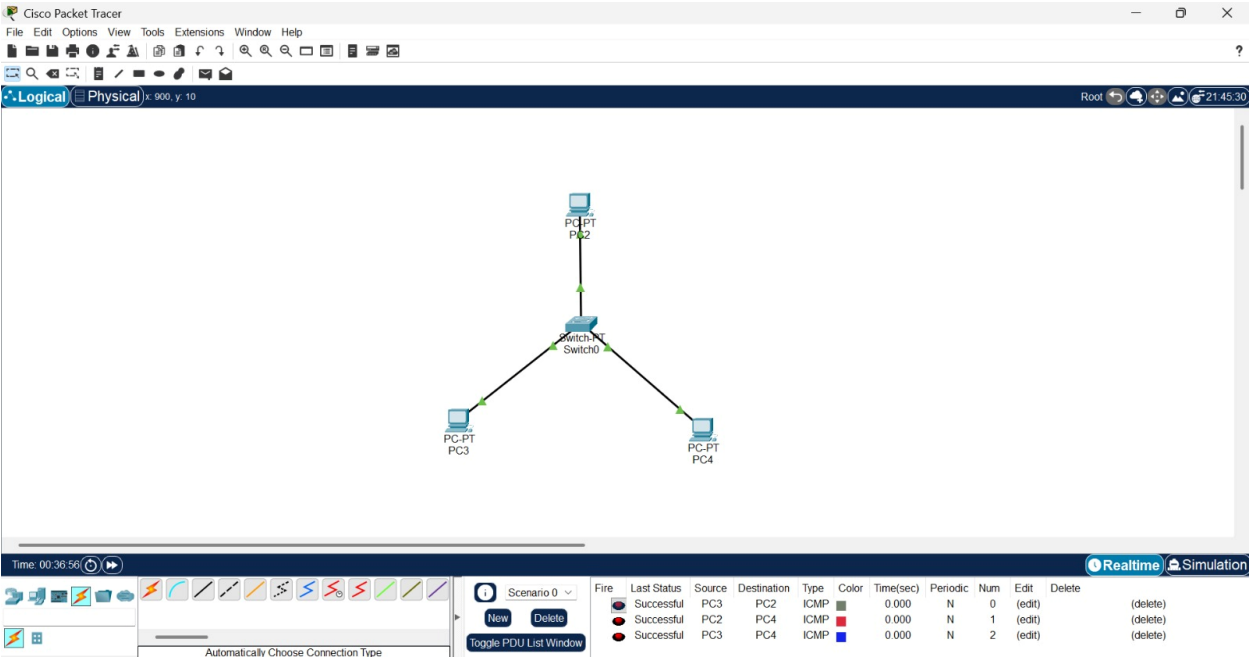
1. Open Cisco Packet Tracer.
2. Place one switch and three PCs in the workspace.
3. Connect each PC to the switch using Copper Straight-Through cables.
4. Configure the IPv4 addresses and subnet masks for all PCs.
5. Open the Command Prompt on PC0.
6. Clear the ARP cache using the arp -d * command.
7. Start packet capture in Wireshark.
8. Execute the ping command from PC0 to PC1.
9. Stop the packet capture.
10. Identify the ARP Request and ARP Reply packets.
11. Analyze the sender and target MAC/IP addresses.
12. Record the observations and verify successful communication.

Algorithm

ARP Operation and Address Resolution Analysis

1. Open Cisco Packet Tracer.
2. Create a LAN with two or more PCs connected to a switch.
3. Assign appropriate IPv4 addresses and subnet masks to all PCs.
4. Verify the IP configuration.
5. Clear the ARP cache on one PC using the arp -d * command.
6. Start packet capture in Wireshark.
7. Execute the ping command from one PC to another.
8. Stop the packet capture after communication is complete.
9. Identify the ARP Request and ARP Reply packets.
10. Analyze the sender and target MAC/IP addresses.
11. Record the observations and explain the ARP resolution process.

OUTPUT:



RESULT:

The ARP Request and ARP Reply packets were successfully captured in Wireshark. After ARP resolution, the ping was successful.

4. **Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.**

Aim

To develop a simple network consisting of two PCs connected through a router using Cisco Packet Tracer, configure the required IP addresses, verify connectivity using the **ping** command, capture ICMP traffic using Wireshark, identify the Echo Request and Echo Reply messages, analyze the ICMP header fields, calculate the Round Trip Time (RTT), and understand the role of ICMP in network troubleshooting.

Objectives

- Design a simple network using a router and two PCs.
- Configure IP addresses and default gateways.
- Verify connectivity using the ping command.
- Capture ICMP packets using Wireshark.
- Analyze ICMP header fields.
- Calculate the Round Trip Time (RTT).
- Understand the role of ICMP in network troubleshooting.

Requirements

- Cisco Packet Tracer
- Wireshark
- One Router (1941/2911)
- Two PCs
- Copper Straight-Through Cables

Theory

The **Internet Control Message Protocol (ICMP)** is a Network Layer protocol used for error reporting, network diagnostics, and connectivity testing. The **ping** utility uses ICMP to verify whether a destination device is reachable.

When the **ping** command is executed:

- The sender transmits an **ICMP Echo Request (Type 8)**.
- The destination replies with an **ICMP Echo Reply (Type 0)**.
- The time taken for the request and reply is called the **Round Trip Time (RTT)**.

Important ICMP fields include:

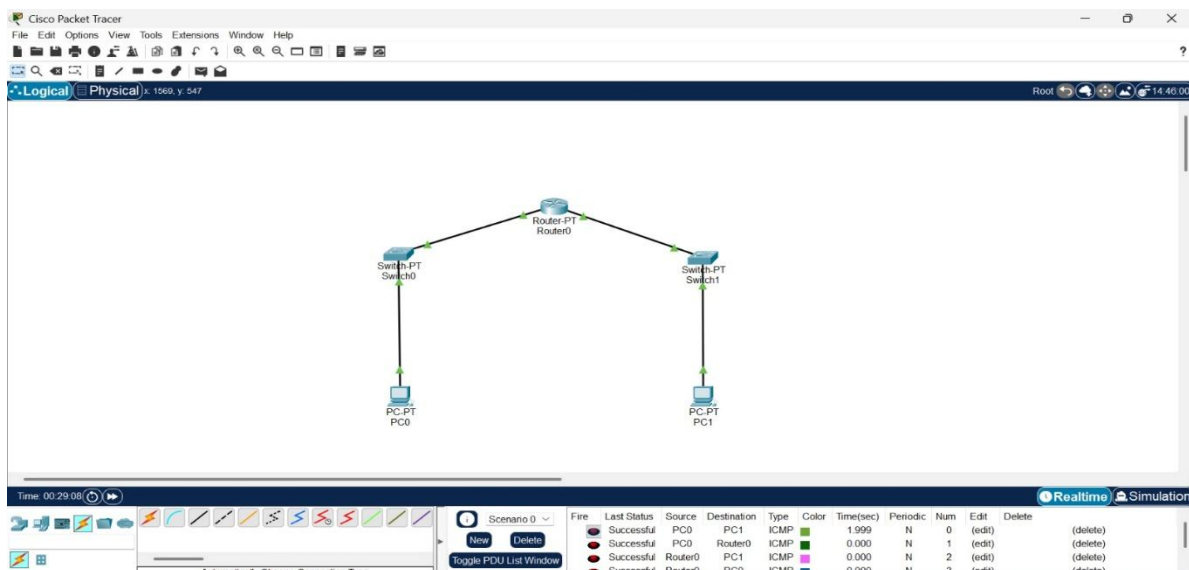
- **Type** – Identifies the ICMP message.
- **Code** – Specifies additional information about the message type.
- **Identifier** – Matches requests with replies.
- **Sequence Number** – Identifies the order of ICMP messages.

ICMP helps network administrators diagnose connectivity issues, routing problems, and network delays.

Algorithm

1. Open Cisco Packet Tracer.
2. Place one router and two PCs in the workspace.
3. Connect each PC to the router using Copper Straight-Through cables.
4. Configure the IP addresses and subnet masks for both PCs.
5. Configure the router interfaces with IP addresses and enable them using the no shutdown command.
6. Configure the default gateway on each PC.
7. Verify connectivity using the ping command.
8. Start packet capture in Wireshark.
9. Execute the ping command again to generate ICMP traffic.
10. Stop the packet capture.
11. Identify the ICMP Echo Request and Echo Reply packets.
12. Analyze the ICMP fields (Type, Code, Identifier, and Sequence Number).
13. Calculate the Round Trip Time (RTT).
14. Record the observations and verify successful communication.

OUTPUT:



RESULT:

The ping is successful, confirming communication between the two PCs through the router. Wireshark captures the ICMP Echo Request and Echo Reply packets, which are used to verify network connectivity.

- 5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.**

Aim

To create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC, enable the HTTP service on the server, access the hosted webpage from the client using a web browser, capture the communication using Wireshark, identify the TCP three-way handshake, HTTP GET request, and HTTP response messages, and analyze how TCP provides reliable delivery for HTTP traffic.

Objectives

- Create a client-server network using Cisco Packet Tracer.
- Configure IP addresses for the client and server.
- Enable the HTTP service on the server.
- Access the hosted webpage from the client.
- Capture TCP and HTTP packets using Wireshark.
- Analyze the TCP three-way handshake.
- Understand how TCP ensures reliable HTTP communication.

Requirements

- Cisco Packet Tracer
- Wireshark
- One Server
- One Client PC
- One Cisco 2960 Switch
- Copper Straight-Through Cables

Theory

The HyperText Transfer Protocol (HTTP) is an Application Layer (Layer 7) protocol used to transfer web pages between a client and a web server. HTTP uses the Transmission Control Protocol (TCP) at the Transport Layer (Layer 4) to provide reliable communication.

Before HTTP data is exchanged, TCP establishes a connection using the Three-Way Handshake:

1. SYN – The client requests a connection with the server.
2. SYN-ACK – The server acknowledges the request and agrees to establish the connection.
3. ACK – The client acknowledges the server's response, completing the connection.

After the connection is established:

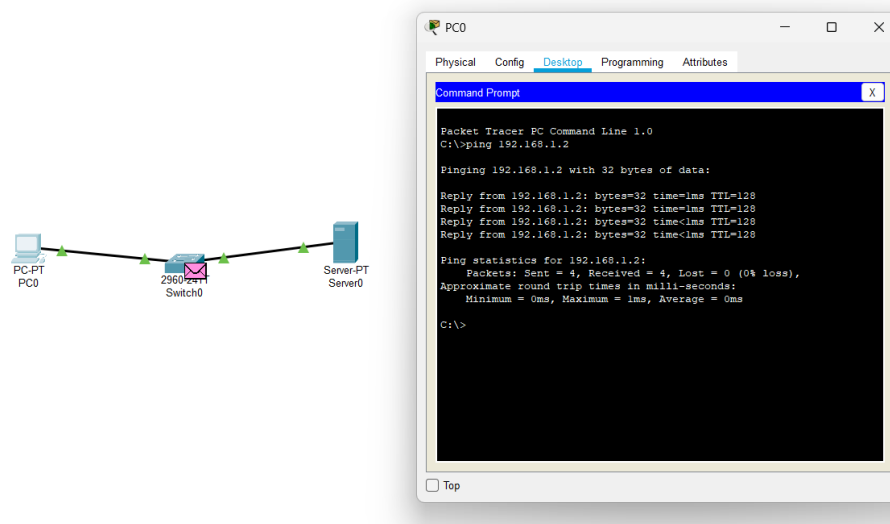
- The client sends an HTTP GET request for a webpage.
- The server returns an HTTP Response containing the requested webpage.
- TCP ensures that data is delivered reliably, in the correct order, and without duplication using sequence numbers and acknowledgments.

Algorithm

1. Open Cisco Packet Tracer.
2. Place one switch, one client PC, and one server in the workspace.
3. Connect the client PC and server to the switch using Copper Straight-Through cables.
4. Configure the IP addresses and subnet masks for both devices.
5. Open the server configuration and enable the HTTP service.

6. Save the default web page on the server.
7. Open the web browser on the client PC.
8. Enter the server's IP address (http://192.168.1.10) and access the webpage.
9. Start packet capture in Wireshark.
10. Refresh the webpage to generate HTTP traffic.
11. Stop the packet capture.
12. Identify the TCP three-way handshake (SYN, SYN-ACK, ACK).
13. Identify the HTTP GET request and HTTP Response packets.
14. Analyze the communication and record the observations.

OUTPUT:



Result

A client-server network was successfully created in Cisco Packet Tracer. The HTTP service was enabled on the server, and the client accessed the hosted webpage through a web browser.

Wireshark captured the TCP three-way handshake, HTTP GET request, and HTTP response, demonstrating reliable HTTP communication using TCP.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation